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The Use of AI in Demand Forecasting for Small and Medium Enterprises

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Abstract

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Having accurate demand forecasting is crucial for SMEs. The current traditional statistical methods used mostly by SMEs often struggle with the lack of high-quality data available. AI techniques can provide more precise forecasts, but currently there are not much guidance available specifically for SMEs.

This study will show how can SMEs adopt AI demand forecasting, and what benefits and barriers they encounter compared with classical approaches. It is based on reviews of the most up-to-date academic papers, industry reports and SME case studies. The available literature was analysed in depth and the performance levels of statistical, AI and hybrid models were compared.

The results show that AI models can significantly decrease forecast inaccuracies, but also highlights that success is linked to data quality, well-planned implementations and external support. The thesis offers a step-by-step framework that provides SME managers and policymakers a practical and straightforward roadmap for an effective AI implementation.

Keywords: AI, supply chain, SCM, forecasting, demand

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Glossary

AI	Artificial Intelligence
B2B	Business to Business
ML	Machine Learning
SEO	Search Engine Optimisation
SME	Small and Medium Enterprise

1 Introduction

1.1 Background and Context

The technological advancements that have been taking place in recent years have had a big impact on the business world. They influence how companies operate and compete with each other. Among them, the most important one has been the development of Artificial Intelligence (AI). When we talk about AI, it refers to computer systems that can learn from different data that gets input to them, they can also recognize patterns and independently make decisions without big human intervention. Nowadays there is a large number of companies that already use AI to make processes more efficient and optimal, such as improvements in customer service, production planning and even sales forecasting (Loureiro, S., Guerreiro, J. and Tussyadiah, I., 2021).

Among the many different areas in which AI could help, demand forecasting is one of the most important ones and could benefit from great improvements. Demand forecasting consists of trying to anticipate the future demand for a product or service. Usually, companies use methods based on historical statistics, but these kinds of methods have difficulties handling aspects like market changes or some difficult data patterns. On the other hand, forecasting based on AI methods has available advanced algorithms like Machine Learning (ML) and neuronal networks, with which big sets of data can be analyzed. These methods can also add to the analysis of things like promotions or seasonality and even update their predictions in real time (Han et al., 2021; Amosu et al., 2024).

Small and Medium Enterprises (SMEs) are significant for every country's economy since they help with the creation of jobs, and local innovation and make the economy grow. Even though they are important, SMEs face challenges that bigger companies do not have. One key challenge is the limited financial and technical resources available for the SMEs. While large

companies can invest in advanced and expensive AI systems, SMEs for their part often use many methods and tools because advanced technology seems too costly or complex. In recent years, there has been growing interest in finding ways to make AI more accessible to SMEs. New models and cloud-based services offer the possibility to use AI tools without a huge upfront investment. Research shows that even simple AI tools can significantly improve demand forecasting accuracy, which in turn can reduce the risks of overstocking or stockouts (Aktepe, Yanik & Ersöz, 2021). As businesses become more digital, it is important to understand both the opportunities and the challenges entailed by the integration of AI into everyday operations.

This study is set in a particular moment of technological development, where there are a considerable amount of technology improvements and the use of AI in business is becoming more common. However, the benefits of these technologies are not available for everyone yet. Large companies got important benefits from their high investments in these technologies, but SMEs still struggle to access them. This thesis will focus on analysing how SMEs can effectively use AI regardless of their resource limitations to improve their demand forecasting.

1.2 Problem Statement

For companies it is important to have a good and accurate demand forecasting procedure in place. For SMEs this is even more important due to their smaller budgets which limit their resources. Some traditional forecasting methods, such as simple averages or some basic statistical models, often fail to consider the different factors that could influence real customer demand. The traditional methods usually depend mainly on historical data and are not good at adapting to sudden changes in market conditions (Aktepe, Yanik & Ersöz 2021).

SMEs do not have high quality and good organized data in their systems in general. Because of the lack of good and reliable data, the AI models have difficulties to analyse it and make accurate predictions with it. As well, the cost of implementing AI systems can be very high for SMEs since they usually do

not have the budget needed for it (Kukreja 2025). Finally, many SME managers and staff members may hesitate or mistrust AI technologies due to a lack of in-house technical expertise. (Foroohar, 2019).

The combination of all these challenges for SMEs makes the potential benefits from AI demand forecasting too accessible. This thesis will study and analyse the challenges that SMEs have and how AI could be implemented successfully for them, taking into account effectiveness and affordability.

1.3 Research Objectives and Questions

This study focuses on how AI can help improve demand forecasting in SMEs. This study will also look at the challenges entailed with using AI in settings with fewer resources and data available, as well as how ready organizations are to adopt AI technologies. The thesis will try to fill a gap in the current research where most studies focus on large companies and expensive AI applications.

Research Question:

What are the challenges and opportunities that AI implementation brings to demand forecasting for SMEs, how can AI be useful in demand forecasting for SMEs?

Subquestions include:

- What are the main challenges (cost, limited data, lack of technical expertise) that prevent SMEs from using AI forecasting methods?
- How do AI forecasting models compete with traditional statistical methods in performance, speed, and ability to handle limited and varying data typical of SMEs?
- How can AI tools, like neural networks and support vector regression, be improved to meet the requirements of SMEs, especially in industries that operate with irregular demand patterns?

- What role do vendor solutions, industry associations, and government policies play in helping SMEs integrate AI into their forecasting practices?

This thesis will help to understand the benefits and limitations of current AI forecasting methods.

1.4 Structure of the Thesis

The structure of this thesis follows a division of six chapters, in which each chapter tries to build on the previous one, and in this way explain how AI can help improve demand forecasting in SMEs.

1 Introduction:

This chapter sets the stage for the study. It explains the background of supply chain challenges, how AI has changed business practices, and the specific issues that SMEs face with demand forecasting.

2. Theoretical Background:

This chapter discusses a more theoretical approach of Supply Chain Management, Demand Forecasting and AI in general terms while relating it to the main objective of this study.

3 Literature Review:

This chapter reviews the existing research from the selected sources. It indicates how AI has evolved its use in business and forecasting models. The review explains why having accurate forecasting is crucial for the success of SMEs and which challenges they face.

4 Research Methodology:

This chapter explains the research design and the methods used for this thesis. The main focus relies on reviewing and analysing researched secondary data, how sources were selected, and their coverage and limitations.

5 Analysis and Discussion:

This chapter analyses all the evidence collected from the literature and presents the main benefits of AI forecasting for SMEs. It highlights the practical barriers to adopting it, and the differences between statistical, AI, and hybrid models. It also discusses the implementations needed in management and policy implications. Here are the foundations for which later will be the proposed framework proposed.

6 Proposed Conceptual Framework:

This chapter introduces a framework by layers that connect several important factors needed for the success in AI implementation. The relationships between them will offer the decision-makers a clear, step-by-step roadmap for implementing AI demand-forecasting.

7 Conclusion:

The final chapter summarises the key findings. It shows AI's potential to improve forecast accuracy and supply chain resilience.

2 Theoretical Background**2.1 Supply Chain Management**

The concept of Supply Chain Management (SCM) usually refers to the coordination of all activities needed to produce and deliver goods or services to customers. It covers the entire chain from the raw materials collection to delivering the finished products to consumers (Kukreja, 2025). SCM puts together key components across the company's value chain like for example:

- Sourcing and procurement, which involves the collection of the raw materials and the inputs from the suppliers.
- Production and operations, which are the processes related to manufacture the goods.

- Distribution and logistics, including warehousing, inventory management, and transportation of the goods to the next figure in the supply chain.
- And demand planning, which focuses on the forecast of the customer demand, which the task to inform procurement and production correctly to make the right inventory decisions.

The correct management of these components is critical for the success of the businesses, including SMEs. A supply chain that is correctly managed can help reduce costs, improve efficiency and increase customer satisfaction overall. Demand forecasting guides at each stage of the supply chain, making it the most important component of the SCM (Aktepe et al., 2021). Having the right forecasting provides critical information for each phase of inventory management and procurement. This is especially important for SMEs, since they often operate with more limited resources and smaller buffers in time and money, therefore the impact of supply demand mismatches can affect them more than larger firms (Kukreja, 2025).

For example, if a SME has too much inventory, it can affect their cash flow, on the other hand running out of stock can affect the customer satisfaction. SMEs should be able to allocate better their resources if they anticipate the demand correctly. However SMEs have limitation as well in technology which affects their access to advance tools. Therefore having access to reliable and advanced demand forecasting methods is critical for their success (Tawil et al., 2023).

In summary, SCM provides the needed framework within which demand forecasting operates correctly. For SMEs having a good forecasting is one of the most important aspects for maintaining a lean, responsive, and competitive supply chain.

2.2 Demand Forecasting Before AI

Before the artificial intelligence techniques became more common, companies usually used traditional statistical methods for their demand forecasting. Some of the common approaches are simple moving averages, exponential smoothing, and autoregressive integrated moving average (ARIMA) models. These traditional methods are in general easy to implement and interpret. They also work well when demand patterns are stable. For example, a moving average or exponential smoothing forecast can correctly get trends in historical sales when market dynamics are steady. Models like ARIMA and exponential smoothing have been for long time in business forecasting practices (Petropoulos *et al.*, 2022). These methods provided a baseline of forecast accuracy and were adopted by SMEs since they have a low cost and do not require too advanced data.

It is important to note that traditional forecasting methods come with some limitations. They often have a difficulties to capture complex or nonlinear patterns in demand and to adapt quickly to sudden market changes. One of their downsides can be explained since they assume that past trends will continue in the future demand based on the historical patterns that they are fed with. Some studies highlight that the traditional methods are not effective managing these drastic changes and irregular demand patterns (Douaioui *et al.*, 2024)

For example, an ARIMA model could provide poor forecast if the consumer preferences change suddenly due to some event or new trend in the market that never happened before. As well, traditional models usually incorporate a limited number of variables, therefore they ignore some important external factors like economic indicators, weather conditions, or even social media trends.

In summary, pre-AI methods such as moving averages and ARIMA have been useful and are still widely used. Their strong points depend more in simple and stable conditions, while their limitations like lack of fast adapting to changes and having a difficult time in capturing complex relationships between variables can

affect greatly their results. Due to these limitations, businesses and researchers have been trying to get to more advanced approaches with the goal to improve their forecast accuracy. This created the base for AI forecasting as a revolutionary solution for the SCM (Douaioui et al., 2024).

2.3 Artificial Intelligence Overview for Forecasting

The advances in AI related to ML provide new techniques for demand forecasting. They tackle many of the limitations of the traditional methods. The ML algorithms are able to interpret patterns from large datasets, this includes for example nonlinear and relationships that are dependant on the context of the market situation they are operating. This capability gives AI forecasting models more flexibility and power in understanding complex real world scenarios. ML has the potential to improve forecasting accuracy and supply chain efficiency (Aamer, A.M. et al, 2021).

ML models are able to process data that contains several variables related to the demand, and be able to update their forecast as new data comes in. This allows for a more dynamic forecasting process, which is key to make accurate and reliable decisions in the supply chain. SMEs can get great advantages from using AI tools since they can help them analyse big data sets, like sales history or online consumer behaviours. The AI methods can provide them insights of the market which would be difficult to analyse with the traditional methods.

Some models have been highlighted in the literature as relevant for AI forecasting. One of them is Artificial neural networks (ANNs), which also includes deep learning variants like recurrent neural networks and long short-term memory (LSTM) networks. They are used to catch complex nonlinear patterns in the demand. A second model worth mentioning is Support vector regression (SVR), this model can find in the data demand critical relationships that traditional methods may overlook, even in if working with nonlinear trends. A study also found that neural network based models and support vector machines are the most used algorithms in demand forecasting research (Aamer, A.M. et al, 2021).

The future of AI usage in SME demand forecasting has a big potential to explore. SMEs can be able to get a strong forecasting accuracy and responsiveness like the large enterprises do. The AI tools are also able to be retrained as much as needed with new information which will allow to react faster to market changes. Some studies already discuss examples of how AI demand forecasting systems are getting introduced for smaller businesses (Douaioui *et al.*, 2024). The rapid grow of analysis tools based on cloud services, which often have an user friendly interface, also contribute to reduce the entry point for the adoption of these technologies for SMEs. In summary, AI methods like neural networks and SVR can provide SMEs with tools to get more accurate, responsive forecasting backed up by complete datasets. This will help to reduce costs, and improve their customer satisfaction.

3 Literature Review

The evolution of AI in business has been looked at closely and analysed, along with the use of AI in demand forecasting and its performance versus traditional forecasting methods. In addition, an analysis of the strengths, weaknesses, and gaps in the current literature shows where this thesis could fit in.

3.1 Gathering Key Studies and Theories

Some research papers have shown how AI use with predictive analytics can help improve the forecasting capabilities for different business uses. Loureiro, Guerreiro, and Tussyadiah (2021) mention in their research that AI is being used on a larger scale in different kinds of operations. The operations extend to various fields such as customer service to supply chain management. AI is changing many aspects of businesses and at the same time increasing further the interest and demand of new studies that look at forecasting optimization capabilities.

In their analysis, Makridakis, Spiliotis, and Assimakopoulos (2018) focused on classical methods and machine-learning approaches for forecasting. They identified that even though machine learning models are good at catching

complex patterns, their performance depends heavily on data quality and the forecasting horizon. Therefore, even though AI can perform better, one has to implement it wisely to be successful with it.

The studies that focus on SMEs add an important point of view. According to Kukreja (2025), SMEs are less likely to use AI technologies since they face unique challenges that bigger companies do not have. The researchers, Gaertner et al. (2024) discuss that not any one model can fit a context. They go deeper into the problem and create an automated forecasting system for SMEs. Amosu et al. (2024) show that while AI forecasting can improve inventory management and customer satisfaction, the actual benefits are closely related to the data selection and the particular industry in which the forecasting takes place.

Aktepe, et al. (2021) discuss how a construction machinery company uses different forecasting techniques tailored to the industry. Their study shows that advanced AI methods, such as neural networks and support vector regression, work better than traditional regression methods, especially when demand is unpredictable or irregular.

Han et al. (2021) explored how AI provides innovation in B2B marketing. Even though their research mainly focuses on marketing, it also shows how AI can handle and analyze large sets of data, which is a technique that can be very useful for forecasting demand.

In the 2019 article, Foroohar explains how technology is disrupting the manufacturing industry. She points out that advanced AI systems bring excitement, but they also come with important challenges like high costs and resistance to change. The kinds of issues that hit SME businesses the hardest.

3.2 Summarizing and Evaluating the Research

Evolution of AI in business:

The journey of AI in business started with simple programs and developed into smart machine learning models. Loureiro et al. (2021) show AI now integrates into many business areas, while Makridakis et al. (2018) reveal that new forecasting methods catch unexpected data twists. But these advanced models demand very high-quality data and computing power which can be perceived as a real challenge for many small companies.

AI in demand forecasting:

In demand forecasting, studies by Aktepe et al. (2021) and Amosu et al. (2024) burst onto the scene, proving that AI tools like neural networks and support vector regression often outdo older methods in accuracy. These studies which are a mix of numbers and insights, show that AI can reduce errors and boost daily operations. To successfully use AI, one should have enough good data and well-built models. Gaertner et al. (2024) even rolled out an automated forecasting system that adjusts to the tight budgets of small and medium businesses.

Challenges and gaps:

Even though many studies show the potential of AI, there are still many more that discuss the limitations that it has. Makridakis et al. (2018) warn that if models are not tuned just right or the data quality is poor, the performance can drop. Kukreja (2025) and Foroohar (2019) make several remarks that many small companies go through difficult situations assimilating AI usage due to the high cost and the technically challenging operation of the AI setups. This shows that there is a gap between the theoretical potential and the actual practical use of AI. Therefore, while AI can improve demand forecasting, there are many things unclear about its cost-effectiveness and the required skill level to use it successfully.

3.3 Connection to the Research Question

The literature that has been reviewed for this thesis, aims to provide a good base of knowledge and information to be able to develop a solid answer to the research question.

Taking a look at some of the studies, like the ones by Loureiro et al. (2021) and Han et al. (2021), they talk about the growing impact of AI in business. Then, Makridakis et al. (2018) and Aktepe et al. (2021), both give some clues and evidence of AI's potential advantages in demand forecasting. On the other hand, Kukreja (2025), Gaertner et al. (2024), and Foroohar (2019), put focus on the challenges that SMEs face, like limited resources and data issues related to lack of quality data.

These studies will be analyzed with a critical point of view to build on their existing knowledge. The focus will be on how AI could be adapted to the SME's particular nature. The findings will be synthesised from all the sources to create a conceptual framework that could help SMEs manage to help with AI implementation, efficiently and affordably.

4 Research Methodology

Here it will be explained how the study for the thesis was planned and executed. The approach used to review the literature, the different types of sources collected and the methods that are used to review the literature. As well as the limits of the study due to the available literature currently existing. The objective is to have a clear and straightforward explanation of how the data was used to answer the research question.

4.1 Research Design and Approach

This thesis uses a qualitative research design which is based on the review of available literature. The decision is to use secondary data from published articles, industrial reports and conference papers instead of collecting new data from experiments or surveys. In this way, it is possible to rely on the academic and professional work done by other researchers to be able to understand better what has already been discussed and studied regarding AI demand forecasting models for SMEs.

The main steps in the research design are:

- **Literature research:** The keywords used to start the research were “AI demand forecasting,” “SME supply chain,” “machine learning forecasting,” and “AI in SMEs” were among the terms searched. Thanks to these keywords, relevant articles and reports were detected for the thesis references.
- **Sorting out the sources:** The focus and criteria for the selection of the sources were related to high-quality sources that can be directly relevant to the field of the research question. The selection consists of academic journals, conference papers, and industry white papers.
- **Revision process:** The abstract was the initial touch and analysis for each source. If the abstract related strongly to the research question then the source was read further. The main points and the findings were annotated and the methods used for each source were taken into consideration to assess the quality of the document.

4.2 Data Collection and Secondary Data Sources

This thesis relies on its findings from secondary data, so the information is collected from verified sources that are already published publically. The data is collected from several types of sources:

This study relies entirely on secondary data collected from published sources. The sources have been classified into three categories: academic journals, industry reports and white papers, and case studies. Each category has its strengths and insights that help build the whole perspective on AI demand forecasting for SMEs.

Academic journals:

- **Amosu et al. (2024)** – Examines how AI-driven demand-forecasting models reduce inventory error and raise customer-service levels, presenting before-and-after metrics from several retail settings. The paper offers quantitative evidence that forecast-accuracy gains translate

directly into lower stock-outs and holding costs, making it highly relevant to SMEs evaluating ROI.

- **Han et al. (2021):** This study is mainly focused on AI in B2B marketing. It provides useful insights into how AI can manage big datasets and detect complex patterns. These insights can apply to forecasting demand too.
- **Hussain & Rizwan (2024)** – This study presents a framework for strategic AI adoption in SMEs. It derives from different case analyses and expert interviews. It highlights leadership alignment, skills development and phased rollouts.
- **Kolková et al. (2022):** This study compares AI forecasting models, both, statistical, and hybrid ones. It talks about how these different approaches perform in SMEs and large enterprises, identifying their respective strengths and weaknesses.
- **Kukreja (2025):** This study focuses on how SMEs can adopt AI to improve supply chain and financial strategies. It discusses the practical challenges that small firms face and discusses which are the opportunities available for using and implementing AI effectively in their businesses.
- **Makridakis, et al. (2018):** This study compares traditional statistical methods with machine learning approaches for forecasting. It highlights both the potential and the limitations of advanced AI techniques, especially regarding data quality and computational requirements.
- **Loureiro et al. (2021):** This study discusses which are the applications of AI in demand forecasting in business environments. It also points out emerging trends and the future research needed in the field.
- **Smyl (2020)** – This study describes a hybrid forecasting approach that mixes exponential smoothing with recurrent neural networks. Its results highlight the performance advantages of hybrid models.

- **Venkataramanan et al. (2024):** This study analyses AI techniques used in sales forecasting in customer relationship management systems. It discusses which are the challenges related to data quality and interpretation of models, which are important for understanding the extended use of AI in forecasting.

Industry report:

- **Foroohar (2019):** This Financial Times article talks about how technologies like AI are getting more presence in traditional manufacturing and supply chain practices. It remarks on the practical difficulties that companies have when they try to implement more advanced technologies, especially from a cost and change management perspective.

Case studies:

- **Aktepe et al. (2021):** This paper presents a case study with a focus on the construction machinery industry. It compares many forecasting methods, to show how AI methods, like neural networks and support vector regression, can perform better than traditional techniques. This is crucial when trying to handle irregular demand patterns in the data.
- **Gaertner, et al. (2024):** This paper presents a case study which tries to develop an automated forecasting pipeline for SMEs. It shows how different forecasting models perform in real scenarios and analyses how important is model validation and selection.
- **Tawil et al. (2023):** This paper provides in-depth case studies of 85 UK SMEs. It maps common difficulties in data-driven decision-making, like as fragmented data and cultural resistance. It provides a great and complete contextual detail on organisational readiness.

These three categories help to collect different perspectives on the diverse landscape of AI forecasting solutions in the form of theoretical insights, practical

observations, and case studies. With this approach, it is possible to create a complete and extensive understanding of AI demand forecasting for SMEs and to highlight areas where there is a need for further research or where practical improvements are still lacking.

4.3 Analytical Framework and Conceptual Model

The analysis of the data collected was done using a thematic approach, which helped identify some common themes and ideas presented in the literature. The process is broken down into several steps:

1. **Reading and taking notes:** After reading each source and taking notes on them. The important points were noted down, like the benefits of AI, the different challenges that SMEs have and the comparisons with other traditional forecasting methods.
2. **Identifying the theme:** With the notes on hand, the recurring themes were identified. The main themes consist of:
 - **Benefits of AI forecasting:** In many sources, it has been highlighted that AI can help provide more accurate forecasts since it can identify complex patterns due to its ability to identify complex patterns in data.
 - **AI adoption challenges:** At different sources, there have been references to challenges like data quality issues, high costs, and lack of technical expertise.
 - **Comparative performance:** The used research has compared many times the performance of the AI methods to traditional forecasting methods, highlighting the advantages and disadvantages.
3. **Comparison of information:** The themes were grouped to compare the findings from them to the other themed sources. For example, Makridakis

et al. (2018) place a strong emphasis on the need for quality data with high quality for AI models to operate well, Kukreja (2025) and Foroohar (2019), on the other hand, discuss the difficulties SMEs have in obtaining such data.

4. Development of conceptual framework: With the combination of the themes with the comparisons, the idea of developing a conceptual model was possible. The model provides the key steps for SMEs to adopt AI demand forecasting successfully:

- **Data preparation:** Have at disposition data that is both, high quality and in sufficient amounts.
- **Extended adoption:** Planning the adoption of AI technologies with a progressive approach. First start with small-scale implementations to assess the benefits of AI, and afterwards go on a full scale.
- **Skills training:** SMEs should train their employees to better understand how to work with and understand the new AI systems.
- **External consulting:** Getting support from external partners like industry associations, and applying for government support projects to ease the transition. Also using products or services provided by other companies.

4.4 Limitations of the Study

There have been some limitations detected while reviewing the literature, like the following ones:

- **Secondary data sources:** This study relies on already published research, meaning that if the sources have any limitations or fall into some biases, then they can have an impact on this thesis's conclusion.

- **Quality of reported data:** The conclusions will also depend on the quality of the data collected in the sources, some of them may not provide the whole perspective on the search question, which may limit the analysis.
- **Fast technology advancement:** Since AI and forecasting methods are evolving very quickly, it may cause some of the literature is not be the most up-to-date regarding recent new advancements or even new challenges.

Despite these limitations, the study provides important insights into the use of AI in demand forecasting for SMEs. The methodology used here lays the groundwork for identifying areas where further research and practical experiments could be beneficial.

5 Analysis and Discussion

5.1 Thematic Analysis of the Literature

Benefits of AI Forecasting

In the literature, there are many key benefits of AI demand forecasting shown for SMEs. Starting with the improvement of forecast accuracy, which can be perceived as better inventory management and more cost-efficient. The traditional forecasting methods mostly rely on limited historical data, which can lead to mismatches between the supply and the demand (Amosu et al., 2024).

AI methods can make an impact, since they can analyze large datasets and understand complex patterns, like non-linear trends and interactions. The studies show that ML models, such as neural networks and support vector regression, have lower errors than basic techniques (Aktepe et al., 2021).

In one case, a neural network model performed better than other approaches in predicting spare parts demand, giving lower error rates and showing the value

of incorporating diverse inputs (e.g. economic indicators, seasonality, etc.) into the forecast (Amosu *et al.*, 2024). With more accuracy, AI forecasting can help businesses avoid unpredictable situations like overstocking or understocking, which gives benefits like lower holding costs, reduced markdowns for excess inventory, and fewer lost sales from stockouts (Amosu *et al.*, 2024).

The improved response rate and agility are also benefits that come with AI models. They can be retrained or updated as frequently as new data is available, this allows SMEs to react faster if the market conditions change. AI forecast predictions can consider real-time data, for example recent sales trends or online customer behavior. With this, SMEs can plan better to anticipate changes in demand. For example, an AI system could detect changes in some customer purchasing patterns or incorporate external information like social media sentiment to adjust forecasts accordingly (Venkataramanan *et al.*, 2024).

This kind of response procedure was mainly available only to large firms that have dedicated analytics teams, but now with the help of AI tools, SMEs also can have access to them. With the dynamic demand forecast, SMEs can reduce the risk of stockouts and overstocks due to unexpected events, like increase in demand due to a viral trend or a decrease in stock from a local disruption (Amosu *et al.*, 2024).

Another aspect to consider is the fact that traditional forecasting might consider only past sales, while the AI models can incorporate different data streams, for example, macroeconomic data, consumer web searches, and even customer reviews to explain better the demand variability. One study showed that by including external factors, such as market trends and promotional campaigns, in a neural network model, the company got more precise forecasts and therefore optimized their inventory levels better than they used to do (Amosu *et al.*, 2024).

By continuing to look for benefits, AI forecasting helps in cost reduction and to get more efficient gains in operations. This is achieved by aligning supply closer to the actual demand, with this SMEs can avoid spending too much capital on

extra unneeded inventory and decrease waste from the products they cannot sell (Gaertner, Lippert & Konigorski, 2024).

Some researchers have documented cases where the implementation of AI forecasting provided huge improvements in key performance metrics, for example with inventory turnover increased and stockout incidents decreased, as well as improving profitability and thus customer service (Amosu *et al.*, 2024).

Having better forecast accuracy translates into fewer emergency actions that companies face, like rush orders or last-minute logistical fixes, which these kinds of measures are often expensive. To summarise the overall findings, it can be said that the literature shows how much AI demand forecasting is a game-changer for SMEs. It improves forecast accuracy, responsiveness, and decision quality, which helps to keep down costs and improves the overall customer experience. These benefits altogether help SMEs, allowing them to get close to the operational efficiency of bigger firms (Kukreja, 2025), (Gaertner *et al.*, 2024).

5.1.1 Challenges in Implementing AI in SMEs

While considering all the benefits that come with AI forecasting, SMEs still face many challenges and barriers when trying to implement it. A challenge that is highlighted many times in the literature is the resource constraint, which can be related to financial and human resources. The implementation of AI systems can require a big upfront investment in software, hardware, and skilled personnel, which many SMEs can find unreachable financially (Kukreja, 2025). The cost of getting an AI solution, or developing one in-house can be challenging. Other factors like the infrastructure needed to run and maintain the AI tool can be a major limitation for SMEs. (Kukreja, 2025).

A critical challenge for SMEs is the lack of technical expertise and skills. This is important to consider since to have an effective use of AI in demand forecasting advanced knowledge in data science and machine learning is required. These

among other skill sets are usually not available at SMEs in-house (Kukreja, 2025).

Big enterprises often have at their disposal data analysts or data scientists who develop and maintain the AI models, but for SMEs that can be considered a luxury. The literature shows repeatedly that many SMEs have issues due to skills gaps. Their employees can have a deep domain knowledge of their industry but more limited experience using advanced analytics tools or AI tools (Gaertner, et al. 2024). This means that SMEs can struggle to build AI forecasting models and interpret and use the results effectively. One study has a strong emphasis on the fact that an automated forecasting pipeline is needed because SMEs usually lack the needed data science expertise to manually experiment with multiple models (Gaertner et al., 2024).

Data challenges are also important since AI models need a lot of historical and external data to be able to work well. But SMEs usually don't have enough data, or their data might be incomplete or inconsistent, or not even digital, for example, sales records stored in spreadsheets. Gaertner et al. (2024) a strong emphasis on how companies use different methods in how they measure and maintain their data. This makes a standardized forecasting solution often difficult. SME's data might lack details like lost sales during stockouts, which will limit AI effectiveness. If SMEs use a limited dataset, the AI models may outperform or even perform worse than other simpler statistical methods (Makridakis, et al., 2018). Makridakis et al. (2018) found that traditional statistical methods sometimes outperformed complex AI models when data was limited and required fewer resources.

The integration with the existing tools used by SMEs may come with challenges. This is because SMEs often use old ERP or more basic accounting systems. But these legacy systems usually have a hard time connecting with new AI forecasting tools (Amosu et al., 2024). To make their systems compatible with the AI systems it will require expensive upgrades and complicated workarounds to make them work. Amosu et al. (2024) remarked that older software isn't well designed for modern AI systems. This forces

businesses to invest in middle software or system updates to be able to benefit from AI systems. However SMEs usually don't have the capacity in their IT teams to handle complex integrations, and without proper integration, the AI forecasts might not directly influence ordering or production, which will strongly limit how useful they can be.

Another challenge to consider are organizational and cultural resistance that SMEs may face. The SMEs' employees might be sceptical or feel to some level threatened by the AI solutions, especially if they have always been handling forecasting manually or by using basic tools. Kukreja (2025) highlights that SMEs often may resist AI because they fear job loss, have an attachment to old methods or simply misunderstand the technology. Commonly, many small businesses rely mostly on intuition and their personal experience rather than on forecasts that are provided by advanced AI solutions. This is especially true when AI models give predictions to managers that they cannot fully understand, like when the results look counterintuitive. While simple forecasting methods (like autoregressive integrated moving averages, also known as ARIMA) are clear and easy to explain, advanced AI models (like neural networks) can be unclear at first glance (Amosu et al., 2024). Without having clear explanations of how forecasts are made, managers might hesitate to rely on them (Amosu et al., 2024). To successfully adopt AI, SMEs must act on their employee's trust in the new technology and build a trust base for everyone in the company to be able to adapt to the technological change and ensure forecasts are understandable.

Data privacy and security issues are also important topics to consider since they can also give some challenges. SMEs also must adhere to regulations like the GDPR in Europe, which dictate strict rules on how businesses must handle personal and customer data (Kukreja, 2025). As an example, if an SME is in the process of implementing AI forecasting might use customer sales data or external consumer data, which may raise compliance questions. It may be difficult to make sure that the data usage is lawful, and that customer privacy is safe. It is especially crucial since non-compliance can lead to important fines and can come along with reputational damage (Kukreja, 2025). Making the

connection between SME's data and cloud-based AI services comes with some cybersecurity concerns. If SMEs don't have properly secured systems in place, they could become targets for cyberattacks (Kukreja, 2025).

In summary, even with all the great advantages that come with AI-driven forecasting, SMEs still face several challenges in making it a reality. The high upfront costs, lack of in-house expertise, data limitations, integration issues, cultural resistance, and regulatory issues, altogether pose a threat to the usage of AI in practice (Kukreja, 2025) (Amosu *et al.*, 2024). Adopting AI is a technical and socio-technical challenge since it requires aligning technology with organizational readiness and support.

5.2 Comparative Evaluation of Forecasting Models

5.2.1 Statistical and AI Approaches

Demand forecasting can be classified into two groups: statistical and AI-based methods. The statistical methods, also referred to as classical forecasting, use tools like moving averages, exponential smoothing (Holt-Winters, for example), and ARIMA models. They are based on statistical theory and assume patterns like stationarity or seasonality in the data (Gaertner, Lippert & Konigorski, 2024). They predict future demand by analysing past trends and random noise. AI methods use various machine learning algorithms, for example, regression trees. They learn patterns from the data and can catch complex and non-linear relationships that other statistical methods might skip.

The literature highlights often the topic of performance versus complexity. Some early comparative studies found that in some standard time series forecasting tasks, the traditional statistical models outperformed several machine learning models in terms of accuracy (Makridaki *et al.*, 2018). In the M3 competition data, a large collection of time series, the simpler methods like exponential smoothing and ARIMA gave more accurate forecast results on average than the ones from machine learning models like neural networks, considering that they are also less technically demanding (Makridakis *et al.*, 2018). This discovery

makes sense partly because many business time series, particularly for SMEs, are short or noisy, which makes it difficult for AI models to generalize well since they need big amounts of reliable data to operate well. One study showed that statistical methods are often more suitable for SMEs because they are easier to apply and demand less computing power and technical expertise (Kolková et al., 2022). That is a reason why SMEs would use these methods, since they can be implemented with easier software, like Excel or basic ERP modules, without the need of having a data science team to set all the required parameters needed by more advanced tools with AI.

The success of forecasting depends a lot on the situation and the context we are working on. AI methods showed that they work better when they have available rich data to use. For example, if an SME has collected for many years data with high frequency and with extra variables like pricing, weather, and web traffic, a machine learning method can use this data better than traditional models and deliver more meaningful results, since they can capture complex demand drivers. There is research that supports this, for example, Aktepe et al. (2021) found that for forecasting construction machinery parts, machine learning was way more accurate than the linear regression approach. When the demand is non-linear or influenced by many factors, the AI methods succeed better at spotting patterns that a simpler model may miss (Venkataramanan et al., 2024).

In summary, it is not so clear the winner between statistical and AI-based forecasting method. It is more about finding what fits best for the specific purpose. On one hand, statistical methods are strong, easy to use, and work well with more limited data or limited resources (Kolková et al., 2022). On the other hand, AI methods are flexible and can handle much more data and understand complex relationships in data. They also often give better accuracy on their results when the usual patterns fail or when a large data is available to work with (Aktepe et al., 2021), (Venkataramanan et al., 2024). Many experts have in common a similar opinion that there is not a method that fits all

situations perfectly (Gaertner et al., 2024). Here is where SMEs should think and assess their data, resources, and goals before selecting a method.

5.2.2 Hybrid and Ensemble Methods

The use of statistical and AI approaches complements each other with their distinctive strengths. That is why many experts now use hybrid methods in which they mix models to improve their forecasts. For example, ARIMA or exponential smoothing can be used to capture linear trends and seasonality and then use a neural network for non-linear patterns. With this, each part focuses on what it does better. The ensemble methods run several models at once and combine their results, using averaging as a more common method. With this the errors can cancel out to some extent and get a more reliable forecast (Venkataramanan et al., 2024).

In the M4 forecasting competition, which took place in 2018, the winning approach was a hybrid model that combined exponential smoothing with a recurrent neural network, a deep learning method, into one framework (Smyl, 2020). Smyl's hybrid method was built by combining the clear and robust nature of exponential smoothing with the pattern-learning power of neural networks (Smyl, 2020). The result generated forecasts that were more accurate than using each method separately. This proved that hybrid models can perform better than pure statistical or pure AI approaches.

Ensemble methods could work well for SMEs. Each model can perform differently because of the access that they have to unique data. Gärtner et al. (2024) built an automated forecasting pipeline for SMEs that tested several models like ARIMA, exponential smoothing, regression trees, neural networks, and an ensemble of these. They found that *“no single model is universally superior; rather, an array of models, when integrated within an automated validation framework, can significantly enhance forecasting accuracy for SMEs”* (Gärtner et al., 2024). They found out that the selection of the best model varied company by company. Their system picks or averages models to perform better than one model alone. The six-model group acted as a backup, if one model

struggled, the others helped. This way, there is no need for an expert to choose the best model, and the forecasts improve. The result gives more robust forecasts, as shown by the superior performance of the ensemble in most scenarios. The recommendation can be that model diversity and validation are crucial to achieve better accuracy (Gaertner et al., 2024).

Hybrid methods can also mix human insight with AI, in this case, they are known as augmented forecasting. Some experts suggest that good practice could be using AI to generate a baseline forecast and then letting human planners adjust for events like planned promotions or policy changes. The result of this approach allows the SME managers to keep applying their market knowledge, so their skills are not wasted in the process. Therefore, it works like an ensemble of machine and human intelligence. Amosu et al. (2024) mention that human oversight is important and should be considered, especially for unexpected events, which will help to overcome the limits that AI may have.

In summary, it can be said that hybrid and ensemble methods combined can provide the advantages of both methods in demand forecasting. They tackle the issue of uncertainty created by the selection of the correct model since they either combine forecasts or integrate models to combine their strengths into one. The literature shows that this kind of approach often gives higher accuracy and more stable performance. This is especially true in situations where different products or periods show different patterns that a single model cannot handle in an optimal way (Gaertner et al., 2024). For SMEs, this can be encouraging since it implies that they need to put all their trust in one forecasting technique. By adopting a hybrid or ensemble approach, through existing tools or solutions that automate this, SMEs can achieve forecasting results similar to the ones from bigger corporations, without needing to become experts in each algorithm.

5.3 Managerial and Policy Implications

5.3.1 Strategies for Incremental Adoption

By taking into consideration all the benefits and challenges discussed so far. It can be said that the best approach for SMEs to adopt AI demand forecasting should be planned incrementally and strategically. Rather than just jumping into the full adoption of AI in their supply chain, a more incremental and phased approach would allow the company to learn, adapt, and build capabilities over time (Hussain & Rizwan, 2024). The selected literature and its practical case studies suggest strategies for making this gradual transition a success.

SMEs can start with small steps with a pilot project using general-purpose AI tools (Hussain et al., 2024), instead of investing in expensive AI systems right away. For example, they can start by trying an open-source forecasting library, like an AutoML (Automated Machine Learning) service, on one of their product lines or operational regions. These kinds of tools are usually low-cost or even free. They do not require advanced coding skills. The goal is to learn and develop AI skills starting from a small scale, before scaling up while building trust in the system (Hussain et al., 2024).

With the further development of the pilot phase, some aspects like the skills gap should be addressed. One approach, without the need to hire external specialists from outside, is to train the current staff. This can be done by sending them to courses on data analytics or training programs on machine learning. Usually, governments and different industry associations offer many training programs which target digital skills for SMEs. Another approach is to collaborate with external experts, meaning having partnerships with academic institutions or consultants to get extra guidance. For some researchers, this is a smart way for SMEs to access AI knowledge and mentorship at a low cost (Kukreja, 2025). Here the SME could get a consultant to help set up an AI forecasting system while its staff shadows the process to learn. With some time, the SME can build its small team proficient in data analysis, this will ensure that it can sustain itself and expand the AI forecasting system afterwards on its own.

Continuing, SMEs should step by step improve their data collection methods and improve their management practices, as part of the plan to adopt AI solutions. They could start by collecting all their sales data into one clean and consistent database or cloud storage. Further, SMEs can start to upgrade their data systems, for example, using a basic cloud database or using an ERP to track better demand drivers like promotions and stockouts. They should begin with what they already have, prove the AI's value on that, and then add more data to help improve the model. This way creates a good roadmap that allows us to build a system ready for data and AI adoption.

Change management is key to success in AI adoption. That is why SMEs should address employees' worries and concerns about AI early on. One strategy is to involve the end-users like demand planners in testing the AI model to build trust. It would be crucial to transmit that AI will support decision-making and handle routine tasks, while staff can focus better on strategy. Having available training can show that the company values its employees' growth (Kukreja, 2025). The leadership should show clearly how AI helps in improving performance, for example showing how it helped cut excess stock by 15% using AI forecasts. This transparency approach also can help build a corporate culture more open to innovation.

The next step could be making the AI model more part of the regular operations. For example, by using it to manage automatic reorder decisions for a certain category of products, The SME could move from forecasting one product line increasingly to covering all, while monitoring the performance during the expansion to evaluate its efficacy. It can be important to also gather feedback from users. If some performance issues are detected, they can stop and check to adapt the model again or check if there is a need for more training data before continuing with the expansion. These recommendations follow the ones mentioned to proceed to more complex, and task-specific AI tools only after being familiar with the simpler ones (Hussain et al., 2024).

In summary, this strategy has four phases in total. First the initiation, then the adoption, followed by the adaptation, and finally the consolidation (Hussain et

al., 2024). By following these steps, the risk of failure can be reduced and investing in advanced AI systems before the company is ready for them. This approach helps SMEs handle better challenges like cost, skills, data, and culture while implementing the tech rollout (Hussain et al., 2024) (Kukreja, 2025).

5.3.2 Role of External Support and Vendor Solutions

SMEs need external support to adopt AI use in demand forecasting. SMEs do not have to work alone in this, and in some cases, they simply cannot. By relying on tech vendors, consultants, government or industry programs, they can get great help from them.

Nowadays, tech vendors and AI providers started to offer solutions designed for SMEs. For example, one trend right now is cloud-based AI (AI as a Service). This means that SMEs do not have to build complex forecasting tools in-house, they can just subscribe to a cloud platform that gives access to these tools on demand. This kind of service model usually is on the form of a pay-as-you-go basis, with this it reduces the initial high costs and the technical complications. Big companies like Amazon Web Services, Google Cloud, and Microsoft Azure provide machine learning APIs and AutoML platforms. SMEs can profit from those services to upload their data and get forecasts in a relatively easy and convenient way (Kukreja, 2025). One important advantage is the fact that they are cloud-based, meaning that the service can scale at the same time as the business needs increase. All this without the need for own servers. (Kukreja, 2025). As a last plus, these kinds of solutions often have easy user-friendly interfaces and integration support, which reduces the need for internal IT development.

Looking for more vendor support solutions, there are ERP systems with built-in forecasting analytics. Gartner et al. (2024) mention that by adding automated forecasting to ERP systems, SMEs can make advanced forecasting possible

(Gaertner et al., 2024). Usually, ERP systems include demand planning dashboards powered by AI and the integration is easy if the forecasting function is part of the ERP they have in place. As well it ensures that forecasts directly support decisions like purchase orders or production plans.

Government and industry can represent an important support for SMEs for their AI adoption. The governments are aware that SMEs are important for the development of the economy, and that most of them need a digital change. One way they can help is through financial help, by offering grants or subsidies to decrease the initial cost of AI implementation (Kukreja, 2025). For example, some programs cover up to 50% of the cost of an AI forecasting system or data analytics training. This kind of support helps reduce the financial risk and promotes trial for SMEs who are willing to improve their business. As well providing tax incentives, like credits or deductions for AI investments or R&D, also can help lower the costs for SMEs (Kukreja, 2025).

The collaboration platforms also fall under external support. Those mainly come from industry associations or local business networks that can create forums for SMEs to share experiences and provide resources. Governments can encourage collaborations like this by supporting programs where SMEs, tech companies, and researchers work together to develop AI solutions in a specific domain (Kukreja, 2025). These kinds of hubs, which promote shared innovation, can allow SMEs to try AI tools in a safer and more controlled environment. Worth mentioning the access to best practices, and the possibility to learn from peers in the same sector.

Finally, external consultants and service providers play a role here too, since there are SMEs that prefer to not develop any part of the solution themselves to avoid all the downsides of handling the implementation on their own. This means that there is room for a growing market of analytics consulting firms or startup solution providers focusing on these SME needs. For SMEs, this approach will mean having ongoing costs. But it converts what would have been a fixed cost, building an internal system, into a flexible service expense, probably more cost-efficient. Starting to use such services can speed up access

to advanced AI forecasting without the risk of having a big development process with all the costs and changes that come with it.

In conclusion, external support and vendor solutions can lower the inconveniences for SMEs to adopt AI in demand forecasting. By using cloud-based AI services, integrated software solutions, financial incentives, training programs, and collaborative networks, SMEs can get the tools, knowledge, and resources that they lack. This means that SME managers must look at and take profit from this external support. For governments, this confirms that investing in digital programs for SMEs is crucial for the future of the economy. When SMEs seek their objectives along with some outside help, they can succeed in adopting AI forecasting more easily and stay competitive.

6 Proposed Conceptual Framework

6.1 Framework Overview

After a close analysis of the benefits, challenges and practices, a conceptual framework can be proposed, which can serve as a base for the adoption of AI demand forecasting for SMEs. It will be composed by using the insights presented in the literature to address the context in which the SMEs operate, meaning their internal capabilities and the potential external support available.

The framework will consist of the following layers:

1. Enablers and preconditions. This layer will outline the which are the important key factors needed to achieve AI adoption in SMEs. For example data readiness, skills and external supports.
2. AI adoption process. Here it will describe the step-by-step process to implement AI forecasting. There will be special attention to highlight how to implement it gradually while handling successfully the challenges that may arise.

3. Outcomes and feedback. The results of the adoption will be shown here. This will be from the point of view of how much the forecasting performance has been improved and the business benefits that it brings.

With this structure, it is possible to demonstrate better that the adoption of AI is a continuous process of learning and improvement, not just a one-time implementation thing.

The framework still is conceptual, which means that it is not a technical solution that can be applied to all SMEs directly. Instead, the framework tries to defy the key components to implement AI successfully, that are relevant and the relations between them needed to make it happen.

In summary, the framework has a holistic view, meaning that combines the technical, organizational and environmental aspects to get the whole picture. The framework will be explained part by part the relations between the different components and how it can be applied to SMEs. The main goal of this conceptual model is to serve as a guide for SME decision-makers and stakeholders like policymakers or consultants and make sure that they address all the important components for the effective adoption of AI in demand forecasting.

6.2 Components and Relationships

In total, there have been detected six different relationships between components, will be presented in this section.

1. Organizational preparation and strategy. This component looks into how ready are SMEs internally, especially in terms of leadership support, company culture, and strategic alignment. Having strong and effective leadership support is important for the success of AI projects (Hussain et al., 2024). Having a clear commitment from leaders and a defined digital strategy can help allocate the resources of the company more effectively and encourage innovation. Also, the SME's willingness to innovate with

their processes and their technological skills play important roles. If an SME has a culture that encourages learning and possesses some analytical skills usually adapts faster to AI, but the ones more resistant to change require more training and changes in management style. It can be clearly said that organizational preparation has a big impact at every stage of AI adoption, and with external support, it can also be improved.

2. Data and technological infrastructure. This component looks at how the IT infrastructure and quality of data operate together. To have successful AI forecasting is important to have accurate, consistent, and detailed historical sales data. Therefore, having ready quality data can directly influence the beginning of AI adoption. For that reason, usually, SMEs that have a big data management system, like an ERP system, are more ready to try implementing AI tools. The data quality has to be good too, if it is poor, the initial efforts should be on improving the data collection methods in place. Technological infrastructure should be backed up with good hardware or cloud services that can support the AI solutions plans. If SMEs have limited IT resources they can overcome them by handing them over to external cloud services vendors (Kukreja, 2025).
3. AI forecasting solution selection. The choice of forecasting methods is crucial in the framework. Choosing between statistical models, AI-based models, or hybrid models is a critical process. The choice depends mostly on the SME's conditions in data availability, staff skills, and the forecasting objectives, and if they prefer quick or rather more accurate solutions. If SMEs have at the beginning limited data and expertise, going towards simpler statistical solutions is more recommended initially, according to Kolková & Ključnikov (2022). This framework allows and encourages us to start simple and slowly but steadily adopt more complete solutions when reaching certain confidence levels in AI.
4. Stages of the adoption process. This refers to the steps SMEs must go through to implement AI forecasting. During the first stage, referred to as the pilot stage, the staff skills have a strong influence on how effectively

the company can learn from initial trials. In the integration stage, it is important to have a strong foundation on the data infrastructure to succeed in connecting the AI forecasting tools with existing SME systems. The following is the expansion stage, in which having external support is highly valuable since it can help improve the currently used AI models or expand their use in the company. This adoption process allows one to step back if needed after evaluating the outcomes from each stage to improve data collection if the pilot shows data issues for example.

5. External support and environment. For an SME having external support can significantly influence positively the AI adoption objectives. This support can be in the form of vendor offerings or government incentives which help reduce financial limitations to move faster into full implementation (Kukreja, 2025). Also should be taken into consideration the pressure from other competitors in the SME segment, which are adopting AI as well. This can motivate businesses to adopt sooner to be ahead of their competitors. External support can help improve the staff skills with training programs and provide expert guidance during the adoption stages.
6. Forecast accuracy and business performance. The last component focuses on the measurable outcomes of AI demand forecasting implementation. These outcomes can be in the form of better forecasting accuracy, reduction of inventory costs, fewer stockouts, and increased profitability. It is important to have a well established feedback loop from these outcomes to the components presented at the beginning. This is to reinforce the investment in AI solutions at the company when results show positive changes. For example, if the SME sees a noticeable improvement in forecast accuracy this may make the management invest more resources in AI implementation. However, if the outcomes are not aligned with the expectations, it is suggested to go through again the initial steps like the quality of the data or the selection of the model to identify if something was not properly prepared.

In summary, the components of the framework are connected between them and have interdependence on each other, which is important for AI adaptation. By understanding these relationships SMEs can prioritize which actions should be taken care of first to make sure that each of their critical areas, like data management, selection of the solution and external resources, and implementation plan, are addressed correctly.

6.3 Application in SME

To show how the conceptual framework can be used, let's apply it to an imaginary small business. In this case is a regional wholesaler of consumer goods to improve demand forecasting using AI, to achieve it they should do the following steps:

1. First, the company will start by assessing how ready they are internally, the organizational support they have and the quality of their data. The leadership is enthusiastic about their situation but they face challenges. The data of the sales is badly organised between multiple spreadsheets, and their IT team does not have experience using AI solutions. Following the framework, the SME decides to focus on solving these problems first. They create unified data storage which will contain two years of sales data and also organize an introductory AI in forecasting training program with the help of an external consultant, in this way, they can improve the skill level of their staff. They established clear goals by agreeing to reduce stockouts by 30%. Here the organization makes sure that the leadership support, the staff skills, and the data quality are all in place.
2. Second, the SME should select a forecasting solution that matches their level of preparation by following the framework. An AI forecasting tool that Integrates with Excel would be ideal. The tool should combine statistical and machine learning methods but have easy usage for the users. Since SMEs usually have a limited knowledge of AI use, they should purchase a cloud-based solution since it will allow them to

outsource complex tasks with the help the vendor expertise, so that they can easily fill internal capability gaps.

3. The next step should be to start a pilot project focusing on a current product line that is experiencing some problems. For example, a beverage, since they often experience overstock or shortages. The SME should start by inputting a few months of sales data into the selected tool to get weekly forecasts. Once they are available some AI-generated forecasts, they compared to the traditional manual forecasts, with that the SME can notice that the AI predictions seem to be more accurate. Some problems may arise here, for example, they could notice that the AI model misses information like promotional influences or a new product, which is why is crucial to include all possible available information to get the more accurate forecast information. In this step, the staff involved will gain confidence and trust in AI since they will experience first-hand the reliability it can provide.
4. The following action is the integration of the AI tool into the regular inventory management of the SME. It is crucial to connect the forecasting system directly to the inventory ordering system. The goal is to automate the inventory decisions. This integration can be eased with the right support from the vendor they have hired. Some weekly meetings should be established to review the forecasts and discuss the actual results, allowing in this way to have human oversight of the process. Any potential scepticism from the warehouse staff would vanish and become supportive as they notice less unpredictable shipments and more stable inventory levels. Steadily, the shift towards regular and data-driven decision-making processes will be set in place for the whole organisation.
5. The fifth step consists of expanding the AI solution to more product categories, especially the ones that have a high value or an unpredictable demand. This should follow the same previous approach of first going with a pilot phase and then gradually integrating each new

category into the system. In the case of products with demand highly dependent on seasonability, they should collect additional data before fully automating their forecasts, sticking in this way to the patience and strategic planning advised by the framework.

6. In the last step, as the SME becomes more proficient with the use of AI forecasting, they look for more improvements in the process by joining a university program sponsored by the government. With support from a partnership like this, the SME can integrate new variables related to regional economic data like unemployment rates or fuel prices, to achieve more accuracy in the forecasts. Besides the accuracy, the partnership also helps enhance the internal capabilities, since the SME staff learn new forecasting techniques and data integration methods.

By following the conceptual framework, this SME could transform its forecasting methods to advanced and improved AI forecasting, in a structured way with little to no risks. The key here is the good preparation of the foundation components like the data quality, the staff training, and the leadership support. Factors like the selection of the right technologies aligned with their capacities, the use of pilot testing from the start to build confidence in them, and a continuous process to improve their system are keys to success. Also, external resources like vendor technologies, consultants, academic partnerships, and government support are very effective supports that can help tackle constraints and promote success for the company.

In a real life scenario, the SME experiences implementing AI solutions can vary. Even if some decide to develop internal expertise or fully outsource forecasting, the framework remains applicable for both. Its main purpose is to remind SMEs to not forget any essential aspects like data quality and staff training, and also encourage them to have a suitable solution which aligns with the real capabilities that the company has. Finally, the SMEs which will follow this step-by-step approach can successfully implement advanced and manageable demand forecasting systems. Those systems will effectively respect their resources and particular organizational setups.

7 Conclusions and Recommendations

The thesis studied the use of AI in demand forecasting for SMEs. From this research, many important findings have been found.

AI forecasting offers clear benefits for SMEs. It helps improve forecast accuracy, which leads to reduced inventory levels, lower costs, and better customer service. AI technology can be quite powerful since it can analyse large amounts of data sets and recognize complex patterns, which other traditional forecasting methods might overlook.

However, SMEs can also find important challenges when trying to adopt AI in their processes. Some of them are limited financial resources, lack of technical expertise, poor data quality, and problems with acceptance within the organization. Therefore, SMEs must address these challenges step-by-step instead of expecting quick and straightforward success.

For SMEs to successfully achieve AI implementation it requires to adopt a well-planned and incremental approach. Starting with pilot initiatives and using simpler tools and then gradually progressing to more complex ones. The use of external support such as cloud-based AI services, vendor solutions, and governmental incentives, can help make AI adoption easier by reducing the SMEs' resource limitations.

SME managers should assess their current forecasting capabilities, and try to address the gaps in data quality, IT infrastructure, and staff training. The selected AI forecasting tools should be integrated seamlessly with the business processes in place. Managers also should be able to translate the forecast improvements into real strategies in their operations to help optimize their supply chains.

Policymakers have an important role in supporting SMEs in their AI adoption goals. They can provide financial incentives, training programs, shared technology services, and a robust digital infrastructure. If they make policies

that help encourage vendor competition, and promote AI solutions targeted to SMEs, it will improve access to advanced forecasting technologies for them.

In summary, SMEs can obtain great benefits from AI forecasting if they adopt it in an organized, strategic, and incremental approach. They should also get support from external resources and policymakers to succeed. By following the presented framework, SMEs can improve their supply chain management and strategic position regardless of their resource limitations.

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